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Bookstore - Final Report

Introduction

This report offers an analysis of the JavaFX-developed Bookstore Application. By enabling interaction between the two main user roles of Owner and Customer, the application facilitates the operations of a digital bookstore. Point-based purchases, customer registration, book management and other user behavior are all supported by the system.

Use-Case Diagram Description & Analysis

The Use-Case shows the interactions between the users of the system and the functionality of the application. The different “single-windows” which appear are also an application of the use-case diagrams.

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Use-case name	Buy Book
Participating Actor	Initiated by Customer
Flow of Events	1. The Customer selects the “Buy Book” option. 2. The system displays available books. 3. The Customer selects a book and confirms the purchase. 4. The system processes payment. 5. The system updates inventory.
Entry Condition	- The Customer is logged into the system. - The Customer has sufficient funds and/or points.
Exit Condition	- The book is added to the Customer’s purchase history. - The Customer’s account is charged - The inventory is updated
Quality Requirements	- Message of confirmation appears within 5 seconds

State Design Pattern Rationale:

The State Design Pattern lets an object change its behavior based on its state, making it ideal for applications with multiple states. In our Bookstore application, the state design pattern was used for the customer's status (being Silver or Gold). It is appropriate because a customer's status can be defined in two distinct states: Silver, for when the customer has less than 1000 points, and Gold, for when the customer has greater than or equal to 1000 points. The customer's behavior also dictates their state, because it changes depending on if they choose to redeem, or save their points. Also, another implication of this pattern was the switching between different screens, as it was a single-window GUI, in which different screens replace the prior screen. This way, an organized system is in place such that if the user logs out, it'll return back to the original state.